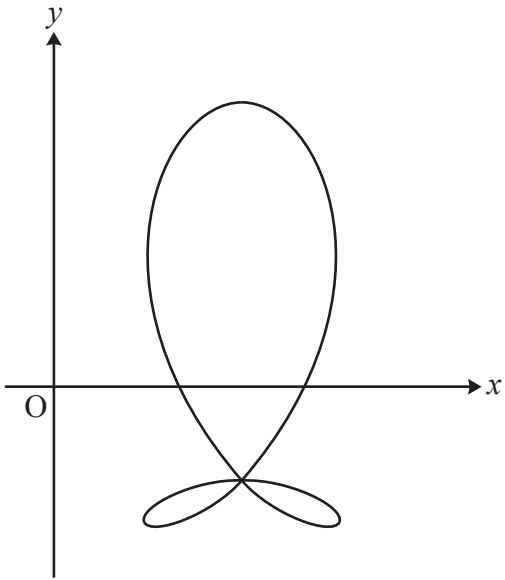
12 The diagram shows the curve with parametric equations

$x = \sin 2\theta + 2,\ y = 2\cos\theta + \cos 2\theta,$ for $0 \leqslant \theta < 2\pi$.



(a) In this question you must show detailed reasoning.

Determine the exact coordinates of all the stationary points on the curve. **[8]**

(b) Write down the equation of the line of symmetry of the curve. **[1]**